Rakey | Ruoqing Yang

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Education

- **University of Nottingham** 2023–2024
  MSc, Human Computer Interaction (graduated with Distinction) *Nottingham, England*
- **Ningbo University** 2018–2022
  BSc, Logistics Management *Ningbo, China*
- **University of Oxford** 2022
  Winter School, Artificial Intelligence and Machine Learning *Oxford, England*

Academic Papers

- **R. Yang** and C. Greenhalgh. "Comparing Experience Intensity of AR and VR for Contrasting Phobia Stimuli." *Virtual Reality*, 2026.
- S. Jin, **R. Yang**, W. Tong, and L. H. Lee. "AIs or Humans in Luxury Shopping? A Design Study of VR Shopping Assistants." *2026 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 1155–1156, 2026.
- S. Jin, **R. Yang**, and L. H. Lee. "Who Should Serve You in Metaverse Luxury Stores? A Comparative Study of 3D Avatars, AI Agents, Webcam Staff, and Voice Assistants." *Electronic Commerce Research and Applications*, under review, 2026.

Academic Experience

- **Comparing VR and AR for Virtual Exposure Therapy** 2024–2025
  MSc Dissertation, Individual Research · University of Nottingham *Nottingham, England*
  - Built the same exposure scenarios in both VR and AR to compare the two media across spider and contamination phobia, following a research-through-design approach.
  - Conducted a mixed-methods study (n=31) with questionnaires, behavioural metrics, and semi-structured interviews, analysing the data through two-way repeated-measures ANOVA and reflexive thematic analysis.
  - Found VR elicited higher experience intensity overall, while AR performed comparably and was more ecologically valid in the spider scenario, pointing to a staged AR-then-VR design.
- **User Perception of Different Shopping Assistants in VR Retail** 2025
  Research Collaboration · University of Cambridge *Cambridge, England*
  - Contributed to the design of a study comparing 4 shopping-assistant modes (3D avatar, webcam, voice, AI) in a luxury VR retail setting, defined the experimental conditions and participant tasks.
  - Co-built the VR retail scenes in Unity and conducted the user study (n=20), running participant sessions and collecting and cleaning the multi-source perceptual and behavioural data.
  - Found 3D avatars to be the optimal assistant mode, while identifying a fundamental realism-immersion trade-off in asymmetric PC-VR interaction.

Professional Experience

- **AI Design Specialist** 01/2024–07/2024
  O Shaped · Internship *London, England*
  - Built AI avatar and generative-video workflows for educational content, cutting production time by 40%.
  - Iterated on the generated outputs from user feedback, improving clarity and usability under real production constraints.
  - Explored digital-human and embodied avatar formats for more present, human-like content delivery.

Projects

- **One's Own Room – AI-Adaptive WebXR Space** *[Github]*
  - Designed a personalised WebXR space adapting to users' affective state to support reflection and restoration.
  - Built an adaptive pipeline combining GenAI 3D world, affect-based atmosphere adjustment, and real-time voice interaction.
- **Virtual Exposure Therapy Application** *[Github]*
  - Built a Unity VR/AR exposure-therapy prototype, presenting spider and contamination scenarios.
  - Used passthrough to switch between AR and VR, with hand-tracking for controller-free interaction. Users trigger object state changes through touch and grab interactions.
- **AR Gallery Treasure Hunt Game** *[Video]*
  - Developed exhibition environment setups and modelled 3D content based on arts exhibit materials and venue geography.
  - Designed location-based AR for spatial exploration, achieving 90% task completion rate.

Awards & Activities

Presented at **GOSIM AI Spotlight, Paris 2026** (Top 10 Frontier Creators), using AI Art to explore AI–human interaction.
Active hackathon participant – **First Prize**, Cambridge EduX Hackathon; **Best AI Use Prize**, RealityX (AI×XR) Hackathon.

Skills & Languages

Research: Mixed Methods (Quant/Qual), Research-through-Design, Experimental Design, User Studies
Data Analysis: Quant – R, SPSS, Python; Qual – Qualitative Coding, Thematic Analysis
XR / Game Dev: Unity (C#), WebXR (Three.js, WebGL), Spark AR **Design:** Figma, Blender, Adobe Creative Suite
Languages: English, Mandarin, Taiwanese Hokkien